GEOMETRY · UNIT 3: TRIANGLE CONGRUENCE

Bridge Truss Inspector

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students inspect a footbridge built from triangles. They decide which congruence shortcuts apply, use corresponding parts to find lengths and angles, work with isosceles triangles, write proofs, and design a truss of their own.

Congruence shortcuts become inspection tools. Students see why SSS makes triangles rigid (the reason trusses work), and why SSA and AAA are not enough.

Standards Addressed

- G-CO.7 Congruence in terms of corresponding parts
- G-CO.8 Explain ASA, SAS, and SSS
- G-CO.10 Prove theorems about triangles (isosceles)
- G-SRT.5 Use congruence to solve problems and prove relationships

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: build a square and a triangle from straws; which one wobbles? Part 1 shortcuts.
Day 2	Part 2 corresponding parts, isosceles triangles, proofs.
Day 3	Part 3 design and report.

Materials

- Student handout
- Straws or craft sticks and pins (optional truss models)
- Rulers and protractors
- Graph paper

How the Three Versions Differ

Level	What changes
CP2	Identify shortcuts from markings, one-variable CPCTC algebra, isosceles base angles, and a fill-in proof.
CP1	Full two-column proofs, HL, CPCTC chains, and isosceles algebra.
Honors	Coordinate proofs of congruence, overlapping-triangle proofs, and a design with an angle constraint.

Watch For (Common Misconceptions)

- Using SSA or AAA as a shortcut.
- Matching vertices in the wrong order in a congruence statement.
- Using CPCTC before proving the triangles congruent.

Extension Ideas

- Build and load-test straw trusses.
- Research Pratt and Howe trusses.
- Compute the angles in the truss with trigonometry (Unit 6).

Answer Key – CP2

Part 1: Inspect the Triangles

#	Task	Answer
1	For each pair, name the shortcut (SSS, SAS, ASA, AAS) or write "not enough information".	Pair 1: SSS. Pair 2: SAS. Pair 3: ASA. Pair 4: not enough information (SSA).
2	Why can't three pairs of congruent angles (AAA) prove two triangles congruent? Draw an example.	Triangles can have the same angles but different sizes (similar, not congruent).
3	In the truss, AB ≅ CD, BC ≅ DE, and AC ≅ CE. Which shortcut proves △ABC ≅ △CDE?	SSS
4	Explain why a triangle frame is rigid but a square frame can wobble.	Three side lengths fix a triangle's shape; four sides of a square can lean into a rhombus.

Part 2: Measure the Parts

#	Task	Answer
5	AB = (3x − 2) ft and DE = (2x + 5) ft. Find x and AB.	3x − 2 = 2x + 5 → x = 7; AB = 19 ft
6	m∠B = (4y + 6)° and m∠E = 62°. Find y.	4y + 6 = 62 → y = 14
7	A truss triangle is isosceles with a vertex angle of 40°. Find each base angle.	(180 − 40) ÷ 2 = 70° each
8	Complete the proof. Given: AB ≅ CB and BD bisects ∠ABC. Prove: △ABD ≅ △CBD.	2. Given. 3. Definition of angle bisector. 4. Reflexive property. 5. SAS.

Part 3: Design a Truss

#	Task	Answer
9	Sketch a truss with at least four triangles. Mark the congruent sides and angles.	Answers vary.
10	Name the shortcut that proves your triangles are congruent.	Answers vary.
11	If every triangle is equilateral with 10-foot sides, how many triangles touch the bottom beam? What is each angle?	40 ÷ 10 = 4 triangles on the bottom beam; each angle is 60°.

Answer Key – CP1

Part 1: Inspect the Triangles

#	Task	Answer
1	Name the shortcut for each pair or explain why there isn't one.	SSS; SAS; ASA; not enough information (SSA)
2	Two right triangles in the bridge's support have congruent hypotenuses and one pair of congruent legs. Which shortcut applies? Why does this not...	HL. It works only for right triangles because the right angle and the Pythagorean Theorem fix the third side.
3	Write a congruence statement for two triangles in the truss and list all six pairs of corresponding parts.	Example: $\triangle ABC \cong \triangle CDE$: $AB \cong CD$, $BC \cong DE$, $AC \cong CE$, $\angle A \cong \angle C$, $\angle B \cong \angle D$, $\angle ACB \cong \angle E$.

Part 2: Proofs & Isosceles Triangles

#	Task	Answer
4	$\triangle ABC \cong \triangle DEF$ with $AB = 3x - 2$, $DE = 2x + 5$, $m\angle C = (5z - 10)°$, and $m\angle F = (3z + 14)°$. Find x, z, AB, and $m\angle C$.	$x = 7$, $AB = 19$; $5z - 10 = 3z + 14 \rightarrow z = 12$, $m\angle C = 50°$
5	An isosceles truss triangle has base angles $(3x + 5)°$ and $(5x - 15)°$. Find all three angles.	$3x + 5 = 5x - 15 \rightarrow x = 10$: base angles 35°, vertex angle 110°
6	Given: B is the midpoint of AD and of CE (segments AD and CE cross at B). Prove: $AC \parallel ED$.	$AB \cong BD$ and $CB \cong BE$ (definition of midpoint); $\angle ABC \cong \angle DBE$ (vertical angles); $\triangle ABC \cong \triangle DBE$ (SAS); $\angle A \cong \angle D$ (CPCTC); these are alternate interior angles, so $AC \parallel ED$.

Part 3: Design a Truss

#	Task	Answer
7	Choose the base length and the height of each triangle. Sketch the truss with markings.	Answers vary.
8	Prove that two adjacent triangles in your truss are congruent.	Answers vary.
9	Find the total length of beam material you need (use the Pythagorean Theorem for the slanted members).	Answers vary. Example: 8-ft bases, 6-ft height: slanted members $\sqrt{(4^2 + 6^2)} \approx 7.21$ ft.

Answer Key – Honors

Part 1: Coordinate Inspection

#	Task	Answer
1	Use the distance formula to show △ABC ≅ △CDE by SSS.	AB = √(3² + 6²) = √45, BC = √45, AC = 6; CD = √45, DE = √45, CE = 6. SSS.
2	Show that △BCD is also congruent to △ABC. Which vertex of △BCD matches A?	BC = √45, CD = √45, BD = 6, so △ABC ≅ △DCB by SSS. A matches D (and B matches C, C matches B).
3	Prove BD ∥ AE using slopes. Then explain how congruent alternate interior angles also prove it.	BD: slope 0; AE: slope 0, so BD ∥ AE. Also ∠DBC ≅ ∠BCA (CPCTC) are alternate interior angles.

Part 2: Proofs

#	Task	Answer
4	Given: B is the midpoint of AD and of CE. Prove: AC ∥ ED.	AB ≅ BD and CB ≅ BE (midpoint); ∠ABC ≅ ∠DBE (vertical); △ABC ≅ △DBE (SAS); ∠A ≅ ∠D (CPCTC); alternate interior angles are congruent, so AC ∥ ED.
5	In the truss, AB ≅ BC, BC ≅ CD, and BD ∥ AC. Prove ∠BAC ≅ ∠BDC.	In △ABC, AB ≅ BC, so ∠BAC ≅ ∠BCA (Isosceles Triangle Theorem). BD ∥ AC, so ∠BCA ≅ ∠CBD (alternate interior angles). In △BCD, BC ≅ CD, so ∠CBD ≅ ∠CDB (Isosceles Triangle Theorem). By the transitive property, ∠BAC ≅ ∠BDC.
6	Explain why SSA fails by constructing two different triangles with sides 10 and 7 and a 40° angle opposite the 7.	Answers vary: both triangles have the 10-unit side and 40° angle, but the 7-unit side can swing to two positions.

Part 3: Design Under Constraints

#	Task	Answer
7	Choose a base length that divides 48 and a height. Show with slopes (rise ÷ run) or a protractor that the base angles meet the rule.	Answers vary. Example: base 8 ft, height 6 ft: base angle with slope 6 ÷ 4 = 1.5 ≈ 56.3°.
8	Write a coordinate proof that two of your triangles are congruent.	Answers vary.
9	Compute the total beam length and explain how the design spreads a load.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Truss design uses congruent triangles with correct markings and a valid congruence justification. Goes beyond the minimum with a creative, realistic choice.	Truss design uses congruent triangles with correct markings and a valid congruence justification.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Inspection report includes congruence decisions with shortcuts, solved measurements, at least one proof, and a labeled truss design. Polished, organized, and visually engaging.	Inspection report includes congruence decisions with shortcuts, solved measurements, at least one proof, and a labeled truss design. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.